

Ruphak Varmaa S

Pre-Final Year CS Undergraduate | Backend Engineering | Distributed Systems | Cloud Infrastructure | AI/ML

+91 7708358305 ruphakv@gmail.com

linkedin.com/in/ruphakvarmaa github.com/RuphakVarmaa ruphak.me

TECHNICAL SKILLS

- **Languages:** Java, Python, C++, JavaScript, TypeScript, SQL
- **Frameworks & Libraries:** React.js, Node.js, FastAPI, LangChain, OpenCV, REST APIs, WebSockets
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, ALB), Microsoft Azure, Docker, GitHub Actions, CI/CD
- **AI/ML:** Machine Learning, Generative AI, Agentic AI, RAG Pipelines, LLM Applications, Computer Vision
- **Databases & Tools:** PostgreSQL, MySQL, Azure SQL, Redis, pgvector, Prometheus, Git, Postman, Linux

EDUCATION

B.E. Computer Science and Engineering (Third Year) 2024 – 2028
KPR Institute of Engineering and Technology, Coimbatore CGPA: **8.5/10**

EXPERIENCE

- AWS Student Builder Campus Leader (AWS)** Jun 2026 – July 2026
- Selected by AWS to lead Builder Center adoption on campus; designed structured onboarding programs across EC2, S3, Lambda, and learning and collaborated with AWS program managers to scale cloud-education initiatives.
- Google Student Ambassador (Google)** Sept 2025 – Feb 2026
- Designed and delivered technical workshops on Cloud Computing and Machine Learning reaching **300+** students; drove adoption of Firebase, TensorFlow, and Google Cloud through live project-based demonstrations.
- Chief Development Officer, Co-Founder (Neural Inverse)** Sept 2022 – Sept 2024
- Architected an automated image-tagging pipeline using Azure Cognitive Services and Azure SQL managing 10,000+ records; led cloud-native backend development and system architecture decisions across all client-facing products.

PROJECTS

- Agentic RAG Document Intelligence System** Python · LangChain · FastAPI · PostgreSQL · pgvector · AWS EC2
- Engineered a multi-agent RAG pipeline using LangChain and GPT-4 for Q&A over 100K-token corpora, achieving 87% retrieval precision via pgvector HNSW indexing with cosine-similarity reranking; reduced manual document analysis time by 70%.
 - Deployed a horizontally-scaled FastAPI service on AWS EC2 behind an ALB sustaining 50+ concurrent sessions at sub-2s p95 latency; implemented adaptive 512-token sliding-window chunking with structured citation metadata for verifiable LLM grounding.
- Real-Time Computer Vision Analytics Pipeline** Python · YOLOv8 · OpenCV · React.js · Prometheus · WebSockets
- Built a sub-30ms inference pipeline fine-tuning YOLOv8 on a 5K-image custom dataset (94% mAP@0.5), streaming bounding-box telemetry over WebSockets to a React.js dashboard rendering per-class histograms at sustained 30 FPS.
 - Designed an asyncio producer-consumer queue decoupling frame capture from model execution, reducing latency variance by 40%; instrumented pipeline stages with Prometheus exposing inference p95, detection rate, and queue-depth SLO alerts.
- Multi-Tenant RBAC Task Management Platform** React.js · Node.js · PostgreSQL · JWT · Azure App Service · Docker
- Architected a multi-tenant SaaS platform with RBAC, JWT authentication, row-level PostgreSQL security policies, and per-tenant audit logging; load-tested to 200+ concurrent users with zero cross-tenant data leakage across 10K+ policy evaluations per session.
 - Reduced query latency from 340ms to 120ms (65% improvement) by rewriting N+1 patterns into batch-loaded joins, adding composite indexes on tenant-scoped columns, and tuning connection pool reuse under burst traffic.
- LLM-Powered Automated Code Review CI Pipeline** Python · LLM APIs · GitHub Actions · Docker · ESLint
- Integrated a GitHub Actions workflow invoking an LLM agent on every PR diff, auto-posting inline comments across 5 severity tiers—OWASP Top-10 security, logic bugs, performance, style, and maintainability—using chain-of-thought prompting for deterministic classification.
 - Automated 80% of routine code review overhead, cutting median PR cycle time by 45% across a 12-engineer repository; containerized the review stage in Docker combining LLM annotations with ESLint and Semgrep static analysis within a sub-3-minute pipeline.

ACHIEVEMENTS & CERTIFICATIONS

- Solved **400+** LeetCode problems (Rating: **1850**) · **Microsoft Azure AZ-900** certified (2024) · **Microsoft Fabric DP-700** certified (2026)